How are the hotels performing?

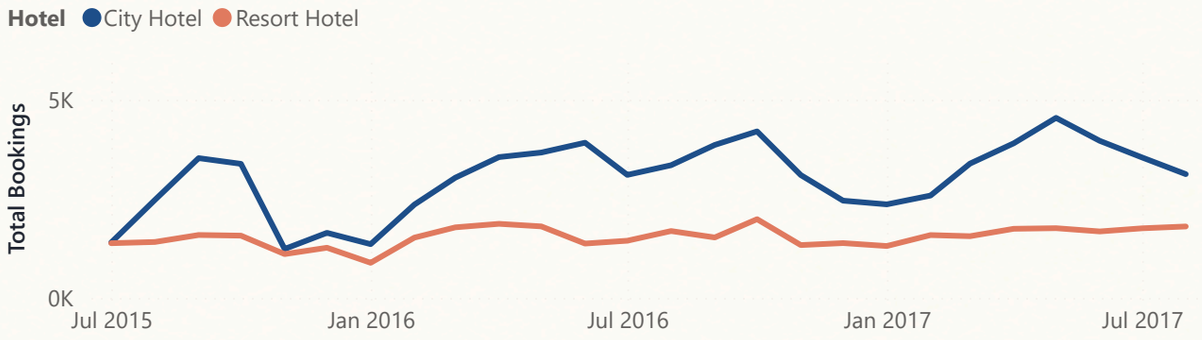
Hotel: All

Date: 1/1/2015 – 12/31/2017

119K	37.1%	€25.95M	€102.24	€16.69M	19.4%	35.1%
Total Bookings	Cancellation Rate	Room Revenue	Avg Daily Rate	Cancelled Value	Bookings YoY %	Room Rev. YoY %

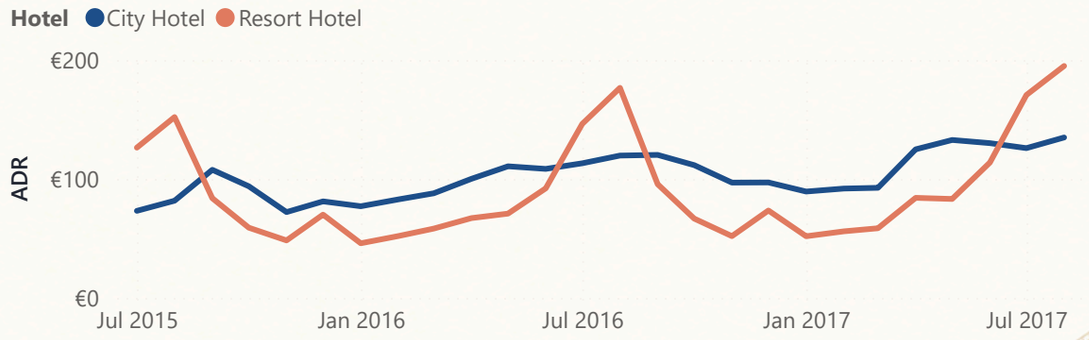
Bookings by month

City Hotel takes about two-thirds of bookings



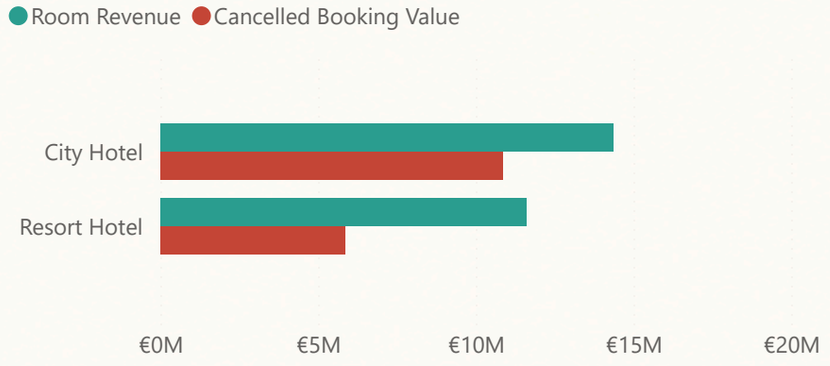
ADR by month

Resort rates rise from about €50 in winter to almost €200 in August 2017



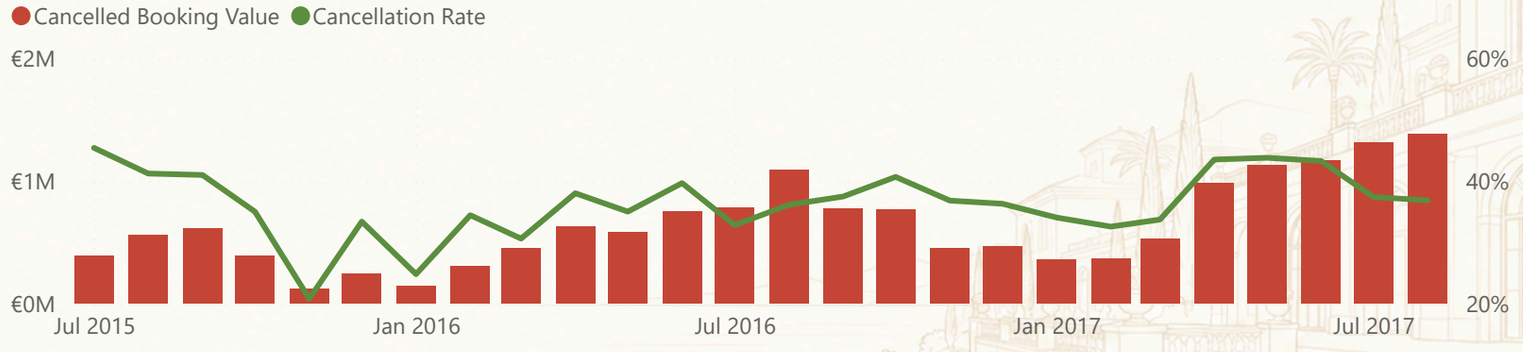
Room revenue vs cancelled value

Cancellations take 43% of City Hotel's booked value, 33% at the Resort



Cancelled value by month

Jan–Aug 2017 was 53% higher than the same months of 2016



Who cancels, and what does it cost?

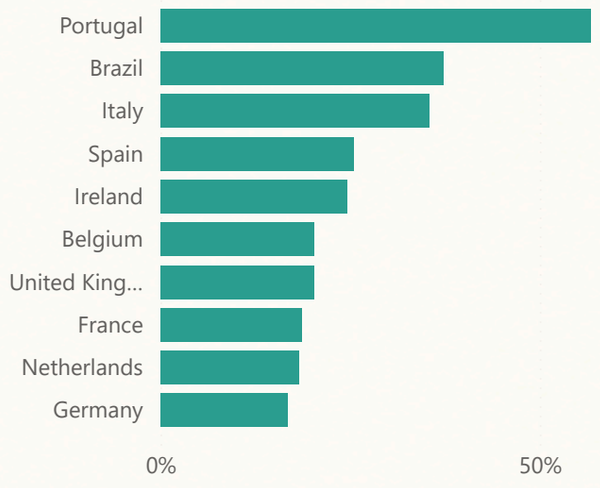
Hotel: All

Date: 1/1/2015 – 12/31/2017

37.1%	€16.69M	39.1%	1203
Cancellation Rate	Cancelled Booking Value	Cancelled Value %	No-Shows

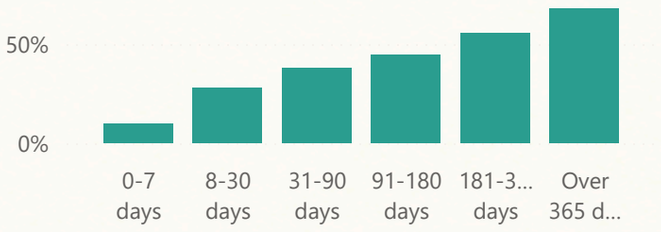
Cancellation rate by country

Top 10 markets by bookings: Portugal cancels at 57%, far above the rest



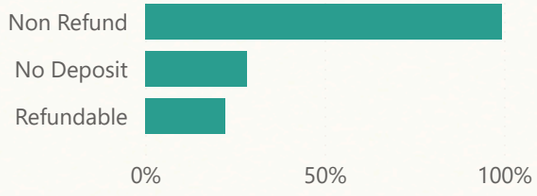
Cancellation rate by lead time

Bookings made more than 6 months ahead cancel more than half the time



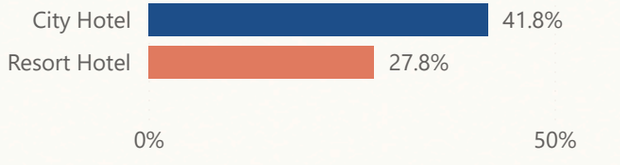
Cancellation rate by deposit type

Non-refundable bookings cancel 99% of the time; 97% are Portuguese, possibly group blocks. Verify before acting.



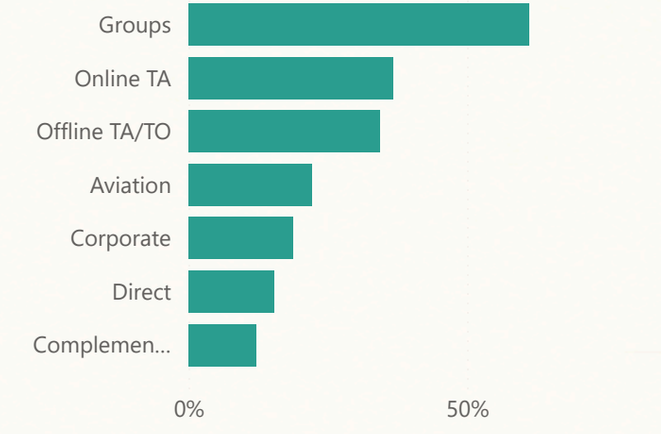
Cancellation rate by hotel

City Hotel cancels more often: about 42% vs 28%



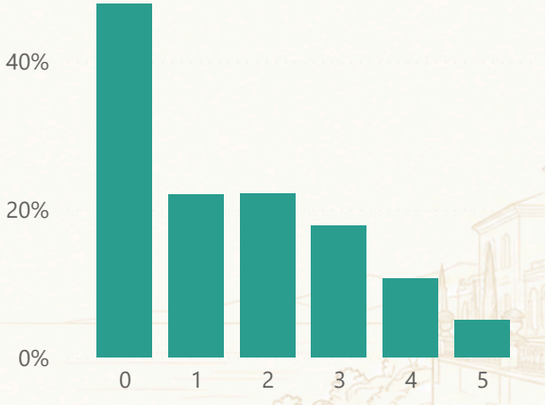
Cancellation rate by market segment

Group bookings cancel most often (61%)



Cancellation rate by special requests

Guests with special requests cancel less than half as often



Key findings

- Bookings made more than 6 months ahead cancel over 55% of the time. A deposit or reconfirmation policy for long-lead bookings is worth testing, but this data can't show it would work: non-refundable bookings here cancel 99% of the time.
- Portuguese bookings cancel at 57%, more than twice the rate of other markets, and make up 97% of non-refundable bookings.
- Guests who make at least one special request cancel less than half as often as those who make none.

Who are the guests?

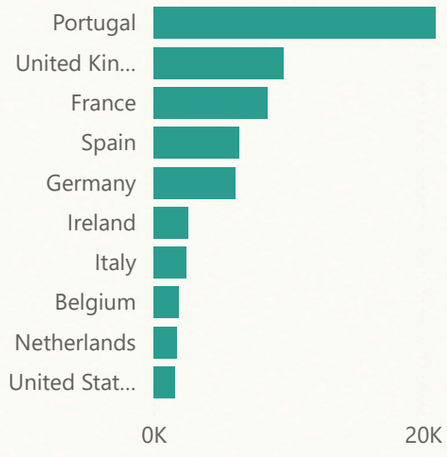
Hotel: All

Date: 1/1/2015 – 12/31/2017

146K	3.39	3.2%	18.6%
Guests Stayed	Avg Length of Stay	Repeat Guest %	Room Change Rate

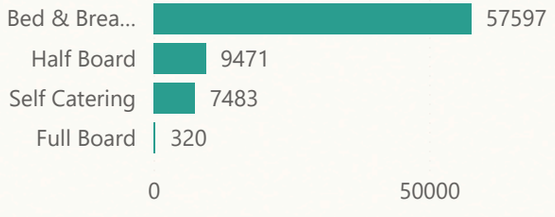
Stayed bookings by country

Portugal, the UK and France are the top guest markets



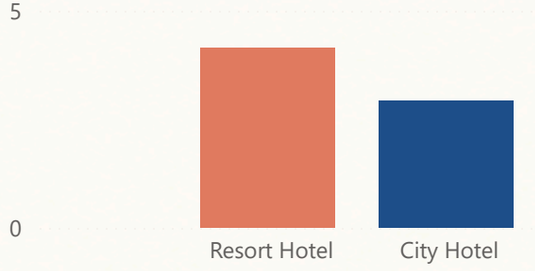
Stayed Bookings by meal plan

Bed & breakfast accounts for about three in four stays



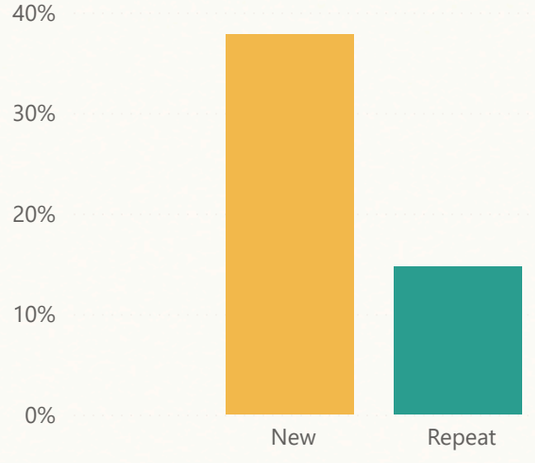
Average length of stay

Resort guests stay longer: 4.1 nights vs 2.9



Cancellation rate: new vs repeat guests

Repeat guests cancel far less: about 15% vs 38%



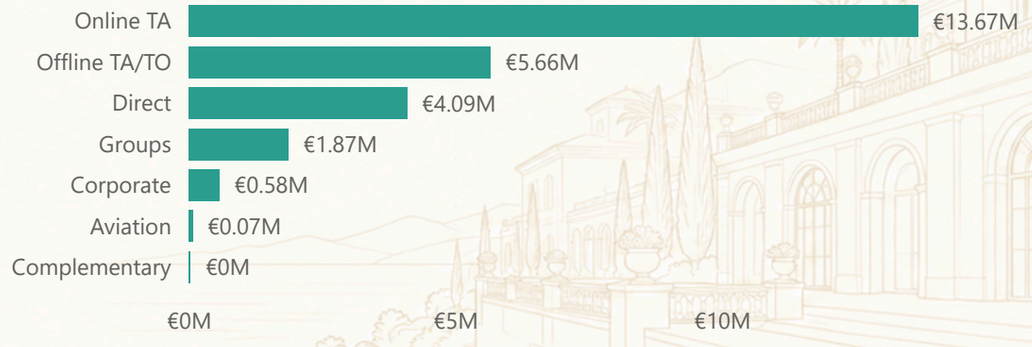
Room type booked vs assigned

Nearly 1 in 5 guests got a different room type than they booked

Room Type	A	B	C	D	E	F	G	H	I	K	Total
A	40631	975	1305	7220	1108	390	177	88	209	135	**52238**
B	83	522		4	2	2	8			1	**622**
C	5	2	577	6	4	2	8	10	9		**623**
D	301	27	32	11689	651	198	79	9	66	25	**13077**
E	11	2	8	21	4036	385	95	3	40	9	**4610**
F	6	12		3	31	1833	111	3	10	3	**2012**
G	4	1	2		4	13	1281	7	14	3	**1329**
H				1			10	339	6		**356**
L	1	1				1		1			**4**
Total	**41042**	**1542**	**1924**	**18944**	**5836**	**2824**	**1769**	**460**	**354**	**176**	**74871**

Room revenue by market segment

Online travel agents bring in 53% of room revenue; groups, which cancel most, bring in 7%



About this report.

Data source

This report analyses 118,985 hotel bookings (119,390 before cleaning) from two hotels in Portugal: a city hotel in Lisbon and a resort hotel in the Algarve. It covers arrivals from 1 July 2015 to 31 August 2017. The data comes directly from the hotels' property management systems. Guest details were removed, and travel agents, companies and room types are shown as anonymous codes.

Source: Antonio, N., de Almeida, A. & Nunes, L. (2019). *Hotel booking demand datasets.* Data in Brief, 22, 41–49. https://doi.org/10.1016/j.dib.2018.11.126

Obtained from the #TidyTuesday project (February 2020), also published on Kaggle as "Hotel booking demand."

Things to keep in mind

- **2015 and 2017 are partial years** (July to December 2015, January to August 2017). Compare the same months across years rather than full-year totals.
- **Non-refundable bookings show a 99% cancellation rate**, the opposite of what you'd expect. It may reflect how the hotels recorded group or block bookings, and it needs checking with the hotels before drawing conclusions.

Room types are letter codes (A to L), because the hotels anonymised them, so room-level findings can't be linked to actual room names. The 12 raw bookings with room type P had no adults and were removed in cleaning.

Definitions

- **Cancelled**: any booking marked as cancelled, including no-shows (guests who never arrived).
- **Stayed**: bookings with the status Check-Out. Revenue, room nights, guest counts and room changes only include stayed bookings.
- **Room nights**: weekend nights plus weeknights.
- **Room revenue**: daily rate × nights.
- **ADR**: room revenue ÷ room nights. All amounts are in euros.
- **Cancelled booking value**: daily rate × nights for cancelled bookings. This is what they were worth, not confirmed lost revenue, as some rooms were likely resold.
- **Cancelled value %**: cancelled booking value ÷ (room revenue + cancelled booking value).
- **YoY %**: change against the same months a year earlier, using only months with data in both years.
- **Lead time**: days between booking and arrival.
- **Repeat guest**: a guest who had stayed at the hotel before. Repeat Guest % is the share of all bookings, including cancelled ones.
- **Room changed**: the guest was given a different room type from the one booked.

Cleaning summary

The raw data (119,390 rows) was cleaned in Power Query. The raw file was never edited.

- Removed **403 bookings with no adults**: 180 with no guests at all and 223 with only children or babies.
- Removed **2 invalid daily rates**: one negative (−6.38) and one extreme outlier (5,400).
- Replaced **4 missing child counts** ("NA") with 0.
- Merged **1,169 "Undefined" meal plans** into Self Catering, as the data documentation says they mean the same thing.
- Replaced "NULL" in **Agent (16,340 rows)** with "No Agent" and in **Company (112,593 rows)** with "No Company". Here NULL means not applicable (no agency used, not a corporate booking) rather than missing.
- Marked **488 missing countries** as Unknown.
- Mapped **CN (1,279 rows) to Canada**. CN is not a valid three-letter code; China already appears as CHN, and Canada's code (CAN) never appears, so Canada is the most likely meaning. This is an assumption.
- Mapped the old code **TMP** to Timor-Leste. Country names and continents come from the international ISO country codes.
- Replaced **"Undefined"** in Market Segment (2 rows) and Distribution Channel (5 rows) with Unknown.
- Combined the separate year, month and day columns into one **arrival date**.
- Added calculated columns: total nights, total guests, lead time bands, room changed, booked via agent, and guest type.
- **Kept 31,976 duplicate rows.** The data has no booking ID, so identical rows may be separate bookings, such as several rooms in one group booking.
- **Kept 645 zero-night bookings**. They add no room nights or revenue, so they don't affect ADR, but they count as bookings and in the cancellation rate.

Final dataset: 118,985 bookings.